UNIVERSITY OF AGRICULTURE

Department of Computer Science

Lab 5: Programming Fundamentals (CS-308)

Basic Information			
Lab #:	5	**Teacher:**	Arsal Mahmood
Objectives	1. For Loop		
Registration#		**Student Name**	
Total Marks	10	**Marks Obtained**	
Tools	Dev C++ / Visual Studio		

Activity 1 : For Loop

1. Calculate the sum of numbers from 1 to 100 and display it on screen by using for loop.
2. Calculate sum of even and off numbers separately from a user given range by using for loop. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and evens in between 50 and 80 and display on screen.
3. Input the marks of a class of ten students. The marks must be between **0** to **100**. Also check marks should not be less than 0 or greater than 100. Calculate the grades of every student and display it by using for loop. Also calculate the total of the grades and the class average.
   - >90 = A
   - <90 and >80 = B
   - <80 and >70 = C
   - < 70 and > 60 = D
   - <60 = F
4. Input a number from user and display a pattern like these on screen by using for loop. For example, user has entered 5, the program should display following pattern on screen.:

   (a)
   ```
   *   *   *   *   *
   *   *   *   *
   *   *   *
   *   *
   *
   ```

P-T-O

(b)

1	2	3	4	5
1	2	3	4	
1	2	3		
1	2			
1				

(c)

1				
1	2			
1	2	3		
1	2	3	4	
1	2	3	4	5

(d)

?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?

(Solve all these questions separately. Also, the pattern would change if the user given number is changed)

5. Calculate sum of even and off numbers by using do-while loop separately from a user given range by using for loop. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and even in between 50 and 80 and display on screen.

6. Print the table of user given number from 1 upto user given limit by using for loop. Like if user enters 3, and limit is 4, program should print table of 3 as follows:

3	*	1	=	03
3	*	2	=	06
3	*	3	=	09
3	*	4	=	12

7. Write a program to find the factorial of a number using a for loop.
8. Write a program that takes an integer and prints its reverse using a for loop.
9. Write a program to check whether a number is a palindrome (same forward and backward) using a for loop.
10. Write a program to calculate x^y (x raised to the power y) using a for loop.
11. Print a diamond shape on screen by using for loop.

```
        *
    *   *   *
  *   *   *   *   *
*   *   *   *   *   *   *
  *   *   *   *   *
    *   *   *
        *
```

{end of Lab}